collection of r independent Wiener processes, then we may write a system of SDEs in $\mathbb{R}^m$ in the form

$$\mathrm{d}\mathbf{Z} = \boldsymbol{a}(\mathbf{Z}, t)\mathrm{d}t + \sum_{i=1}^{r} \boldsymbol{b}^{(i)}(\mathbf{Z}, t)\,\mathrm{dW}_i. \tag{6.15}$$

alternatively, if $\boldsymbol{B}$ is the matrix with columns $\boldsymbol{b}^{(1)}, \boldsymbol{b}^{(2)}, \ldots, \boldsymbol{b}^{(r)}$, and $\mathbf{W}$ represents the r-component vector of Wiener processes, then

$$\mathrm{d}\mathbf{Z} = \boldsymbol{a}(\mathbf{Z}, t)\mathrm{d}t + \boldsymbol{B}(\mathbf{Z}, t)\,\mathrm{d}\mathbf{W}. \tag{6.16}$$

6.3.1 The Itō Formula

We shall derive, heuristically, a core piece of mathematical apparatus which permits much of the analysis of SDEs. Let $\phi(\cdot)$ be any continuously twice differentiable function and let $\mathrm{X}_n \in \mathbb{R}$ represent the state of a biased random walk as considered in Sect. 6.3, i.e., with state $\mathrm{X}_{n+1} = \mathrm{X}_n + a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x$. Let $\phi_n = \phi(\mathrm{X}_n)$, then Taylor's theorem gives

$$\begin{aligned}
\delta\phi &\stackrel{\text{def}}{=} \phi_{n+1} - \phi_n \\
&= \phi'(\mathrm{X}_n)\delta\mathrm{X}_n + \frac{1}{2}\phi''(\mathrm{X}_n)\delta\mathrm{X}_n^2 + o(\delta\mathrm{X}_n^2) \\
&= \phi'(\mathrm{X}_n)(a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x) \\
&\qquad + \frac{1}{2}\phi''(\mathrm{X}_n)(a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x)^2 + o(\delta\mathrm{X}_n^2).
\end{aligned}$$

Now we expand the 2nd order term and, working in the diffusion limit, recognize that $\delta x^2 \sim \delta t$, thus we have

$$\begin{aligned}
\delta\phi &= \phi'(\mathrm{X}_n)(a(\mathrm{X}_n, n\delta t)\delta t + b(\mathrm{X}_n, n\delta t)\mathrm{J}_n\delta x) \\
&\qquad + \frac{1}{2}\phi''(\mathrm{X}_n)b(\mathrm{X}_n, n\delta t)^2\delta t + \mathcal{O}(\delta t^{3/2}).
\end{aligned}$$

Passing to the limit, this leads to the SDE:

$$\mathrm{d}\phi(\mathrm{X}) = \phi'(\mathrm{X})(a(\mathrm{X}, t)\mathrm{d}t + b(\mathrm{X}, t)\mathrm{dW}) + \tfrac{1}{2}\phi''(\mathrm{X})b(\mathrm{X}, t)^2\mathrm{d}t. \tag{6.17}$$